

**Started on** Friday, 14 August 2026, 4:17 PM**State** Finished**Completed on** Friday, 14 August 2026, 4:23 PM**Time taken** 6 mins 4 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)**Question 1** | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.**Input:**

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 #include<stdio.h>
2 int main(){
3     int n;
4     int count=0;
5     scanf("%d",&n);
6     for(int i=n/2; i<n; i++){count++;
7         for(int j=1; j<n; j = 2 * j){count++;
8             for(int k=1; k<n; k = k * 2){
9                 count++;
10                count++;
11            }count++;
12        }count++;
13    }count++;
14    count++;
15    printf("%d",count);
16 }
17
18
19
20
```

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	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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